SIMATS ENGINEERING

Department of Computer Science and Engineering

CSA15 Cloud Computing and Big Data Analytics

CO2 AT1 - Capstone Infrastructure Sizing Workbook

Guided calibration workbook for VM sizing, virtualization decisions, snapshot planning and VM-container comparison

Assessment Metadata

Course Code	CSA1512
Course Title	Cloud Computing and Big Data Analytics
Assessment	CO2 AT1 - Virtualization Scenario Problem-Solving
Submission Type	Individual digital workbook based on the same capstone title used in CO1
Faculty	Dr. C. Rajesh Babu
Employee ID	SSETSCS-415
Submission Mode	Completed workbook and evidence repository link through Google Classroom

How This Workbook Works

- Read each guidance block and immediately fill the related response table.
- Use the same capstone title selected for CO1.
- Make reasonable assumptions when exact values are unknown. Good assumptions must be written clearly.
- The goal is not to guess a perfect industry configuration; the goal is to reason technically and justify the first infrastructure plan.
- The final decisions from this workbook will be used again in CO2 AT2 and CO2 AT3.

Technical Tip

A good infrastructure estimate is transparent. If a number is assumed, write why it is assumed. If a number is calculated, show the formula. If a service is selected, justify why it fits the workload.

Student and Capstone Details

Student Name	M.GUNA SHEKAR
Register Number	192472345
Class / Slot	Slot B
Capstone Title	GreenHire: Cloud-Based Energy-Efficient Resume-to-Job Matcher Using NLP Skill Gap Intelligence
Capstone Product Type	Web app / Android app / iOS app / Hybrid app

CO1 Architecture Reference	Mention concept map / presentation title / GitHub link if available

Activity 1: Bring Forward CO1 Capstone Context

Start with the cloud product idea already used in CO1. The infrastructure plan must support that same application, not a new unrelated example.

Capstone Element	Student Entry
Primary users	Job Seekers, Recruiters, HR Managers
User roles	Admin, Job Seeker, Recruiter
Main modules	Resume Upload, NLP Skill Extraction, Job Matching, Skill Gap Analysis, Dashboard
Cloud services identified in CO1	Virtual Machine, Cloud Storage, Cloud Database, NLP Processing Service
API/backend need	REST API developed using Flask
Database need	MySQL Database
File upload/storage need	Resume (PDF/DOCX) and Job Description Storage
Analytics/AI/Python module	NLP Skill Extraction, Resume Matching and Skill Gap Intelligenc

Activity 2: Workload Classification

Classify the capstone workload before selecting a VM. A form-based SaaS app and an AI analytics app should not receive the same infrastructure plan.

Workload Type	Typical Signals	Starting VM Guidance
Light web / form app	Login, CRUD, simple reports, low file upload	1-2 vCPU, 2-4 GB RAM
API backend	Authentication, REST APIs, moderate concurrent requests	2 vCPU, 4-8 GB RAM
Database-heavy app	Search, many records, frequent writes	2-4 vCPU, 8-16 GB RAM
File-heavy app	Images, PDFs, uploads, downloads	2 vCPU, 4-8 GB RAM + separate object storage
Analytics app	Dashboards, aggregation, batch reports	4 vCPU, 16 GB RAM
AI/Python module	Prediction, NLP, image/text processing	4-8 vCPU, 16-32 GB RAM; GPU optional
High traffic app	Large user base and peak demand	Multiple VMs or autoscaling group

- **Student Decision: My capstone workload type and reasonUses NLP for resume analysis.**
- **Performs skill extraction and job matching.**
- **Processes uploaded documents.**
- **Generates skill gap reports.**
- **Provides analytics dashboard**

Activity 3: User and Traffic Calibration

Use the following formulas to convert user assumptions into approximate traffic.

Metric	Formula	Example
Concurrent Active Users	Registered Users x Active User % x Peak Factor	500 x 10% x 2 = 100
Requests per Minute	Concurrent Active Users x Actions per User per Minute	100 x 2 = 200
Requests per Second	Requests per Minute / 60	200 / 60 = 3.33 RPS
Daily Requests	Daily Active Users x Avg Actions per Day	200 x 30 = 6000

Input	Value	Reason / Assumption
Registered users expected	2000	Medium-scale recruitment platform
Active user percentage	15%	Regular daily users
Peak factor	2	Peak usage during working hours
Actions per user per minute	2	Resume upload and job search
Calculated RPS	20 RPS	(2000 × 15% × 2 × 2) ÷ 60
Daily active users	300	15% of registered users
Daily requests	9000	300 × 30 actions/day

Activity 4: CPU Calibration

CPU sizing depends on request rate and processing complexity. Use the score below as a guided estimate.

CPU Driver	Score
Base web/API application	1
Authentication and role management	+1
Moderate API traffic above 5 RPS	+1
Database-heavy search or reporting	+1 to +2
Background jobs or notifications	+1
Analytics or dashboards	+2
AI/Python inference inside server	+2 to +4

Total CPU Score	Recommended vCPU
1-2	1-2 vCPU

3-4	2 vCPU
5-6	4 vCPU
7-9	4-8 vCPU or separate analytics/AI service
10+	Scale out with multiple VMs or managed compute

CPU Driver Selected	Score	Reason
Base web/API application	1	Flask backend
Authentication and roles	1	Secure login
API traffic	1	Moderate API requests
Database/search/reporting	2	Resume and job search queries
Background jobs	1	Resume indexing and notifications
Analytics/dashboard	2	Recruiter dashboard
AI/Python module	3	NLP skill extraction and matching
Total CPU Score	11	
Recommended vCPU	**8 vCPU**	

Activity 5: RAM Calibration

RAM must include operating system, runtime, application process, database/cache and a safety buffer.

Component	Guidance	Student Estimate
OS + runtime	1-2 GB	2GB
Web/API service	1-2 GB	2GB
Small database	2-4 GB	4GB
Medium database/cache	4-8 GB	4GB
Analytics/reporting	4-8 GB extra	8GB
AI/Python module	4-16 GB extra	8GB
25% buffer	Subtotal x 0.25	5GB
Final RAM	Round up to standard size	32GB

Activity 6: Storage Calibration

Storage must cover application files, database records, uploads, logs and backup/growth buffer.

Storage Item	Formula / Guidance	Student Estimate

OS and application	20-40 GB	40GB
Database	Avg record KB x records/day x days x 1.3 / 1024 MB	50GB
File uploads	Avg file MB x files/day x days x 1.3	80GB
Logs	Requests/day x log KB x days / 1024 MB	10GB
Backup/growth buffer	Subtotal x 30%	54GB
Final storage	Round up to standard disk size	256GB

Technical Tip

Do not store large uploads permanently inside the VM disk when object storage is available. VM disk is for OS and application. Object storage is better for images, PDFs, media and documents.

Activity 7: Select the VM Plan

Plan	When It Fits	Student Choice
Single VM prototype	Small demo, simple app, low traffic	NO
Two-tier design	App VM + managed/separate database	NO
Three-tier design	Frontend, backend/API and database separated	YES
Containerized backend	Microservices, portable API, CI/CD	YES
Serverless function	Event-based task, notification, small automation	NO
Managed database/storage	Reliability, backup and scaling required	YES

Final VM Plan: selected plan, vCPU, RAM, storage and reason

- **Architecture: Three-tier architecture**
- **vCPU: 8**
- **RAM: 32 GB**
- **Storage: 256 GB SSD**
- **Reason: Supports NLP processing, document uploads, scalable APIs, and dashboard analytics while ensuring reliable performance.**

Activity 8: Hypervisor and Virtualization Decision

Approach	Meaning	Best Use
Type 1 hypervisor	Runs directly on physical hardware	Production data center and cloud provider infrastructure
Type 2 hypervisor	Runs above an existing OS	Student lab, local testing, VirtualBox/VMware Workstation

Cloud-managed virtualization	Cloud provider hides hypervisor details	Most public cloud VM deployments

Container runtime	OS-level process isolation	Lightweight APIs, microservices, fast deployment

Context	Selected Approach	Justification
Local prototype/testing	Type-2 Hypervisor	Suitable for development and testing using VirtualBox/VMware
Cloud pilot deployment	Cloud-Managed Virtualization	Easy cloud deployment with managed infrastructure
Production growth scenario	Type-1 Hypervisor	High performance, security, and scalability
Module suitable for container	Backend API (Flask)	Easy deployment, portability, and scalability using Docker

Activity 9: Snapshot and Clone Strategy

Snapshot protects a point in time. Clone creates a duplicate environment. Both are useful, but they are not the same.

Event	Snapshot or Clone?	Frequency / Timing	Retention / Reason
Before major deployment	Snapshot	Before every release	Rollback if deployment fails
Before database schema change	Snapshot	Before DB updates	Protect database integrity
Before OS/package update	Snapshot	Monthly	Recover from update failures
Before review/demo	Snapshot	Before presentations	Stable demonstration environment
Create testing environment	CLONE	When required	Independent testing
Create team demo copy	clone	Every sprint	Team demonstrations

Activity 10: VM vs Container Decision

Criteria	Virtual Machine	Container
Isolation	Strong OS-level isolation	Process-level isolation
Startup	Slower	Faster
Resource use	Higher	Lower
Best for	Full OS, legacy app, database VM	Microservices, APIs, portable services
Portability	Moderate	High
Student use	Clear infrastructure learning	Modern deployment learning

Capstone Module	VM / Container / Managed Service	Reason

Frontend	Container	Lightweight and scalable
Backend/API	Container	Portable Flask application
Database	Managed Service	High availability and automatic backup
File storage	Managed Object Storage	Efficient storage for resumes

AI/Python module	VM	Requires higher CPU and memory
Analytics/dashboard	Container	Easy scaling and deployment
Notification/background jobs	Container	Lightweight background processing

Activity 11: Final Recommendation

Write the final infrastructure recommendation in 8-12 technical lines. Include VM size, hypervisor/virtualization approach, storage plan, snapshot/clone plan and VM-container decision.

GreenHire is designed using a three-tier cloud architecture consisting of frontend, backend, and managed database services. The backend API is containerized using Docker and deployed on a virtual machine with 8 vCPUs, 32 GB RAM, and 256 GB SSD storage. Resume files are stored in managed object storage, while a managed MySQL database stores user profiles, job listings, and matching results. NLP-based skill extraction and skill gap intelligence are executed on the application server to provide accurate resume-to-job matching. Cloud-managed virtualization is recommended for deployment, while Type-2 virtualization is suitable for local development. Snapshots are created before deployments and system updates for quick rollback, and clones are used for testing and demonstration environments. Containerization improves scalability, portability, and resource efficiency. This infrastructure provides a reliable, secure, and energy-efficient cloud platform for intelligent recruitment and resume matching.

Carry-Forward to CO2 AT2 and CO2 AT3

Output from CO2 AT1	How It Will Be Used Later
VM sizing values	Used as configuration target in CO2 AT2 practical evidence
Hypervisor/virtualization choice	Used to justify deployment environment
Snapshot/clone plan	Used as backup and testing evidence
VM/container comparison	Used in CO2 AT3 infrastructure design case
Final recommendation	Used as the capstone infrastructure baseline

GitHub Submission Checklist

- Completed workbook file.
- README with capstone title, sizing summary and final recommendation.
- Optional architecture diagram or table exported as image/PDF.

- Repository link submitted in Google Classroom.